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United States Patent	12414160
Kind Code	B2
Date of Patent	September 09, 2025
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2-step random access method and apparatus, terminal and storage medium

Abstract

The present disclosure discloses a 2-step random access method and apparatus, a device, and a medium, and relates to a field of wireless communication. The method includes: transmitting a message A, when a 2-step random access is triggered based on a trigger event, the message A carrying a Cell-Radio Network Temporary Identifier (C-RNTI) of a User Equipment (UE); receiving a message B; and determining that a conflict of the 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event. The present disclosure may determine, based on whether the message B satisfies the condition corresponding to a trigger event, whether a conflict on a 2-step random access channel is resolved.

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Appl. No.: 17/562555

Filed: December 27, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220124825 A1	Apr. 21, 2022

Related U.S. Application Data

continuation parent-doc WO PCT/CN2019/100872 20190815 PENDING child-doc US 17562555

Publication Classification

Int. Cl.: H04W74/0836 (20240101); H04W74/0833 (20240101)

U.S. Cl.:

CPC H04W74/0836 (20240101); H04W74/0833 (20130101)

Field of Classification Search

CPC: H04W (74/0833)

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Background/Summary

(1) The present application is a continuation of International Application No. PCT/CN2019/100872, filed on Aug. 15, 2019, the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a field of mobile communication, and more particularly, to a 2-step random access method and apparatus, a terminal and a storage medium.

BACKGROUND

(2) A Random Access Channel (RACH) is an important channel in an initial access process between an access network device and a User Equipment (UE). Long-Term Evolution (LTE) adopts a contention-based 4-step random access mechanism, which will be simplified to a contention-based 2-step random access mechanism in some usage scenarios of a New Radio (NR) system.

(3) Regarding the contention-based 2-step random access mechanism, the UE directly transmits a random access preamble and identity information to the access network device, and the access network device transmits an approach for resolving a conflict on the RACH along with a Physical Downlink Control Channel (PDCCH) scrambled by a Random Access-Radio Network Temporary Identifier (RA-RNTI) to the UE, so as to resolve the conflict on the RACH.

(4) However, there are a number of trigger events for a conflict resolution approach for a contention-based 2-step random access. Scenarios corresponding to different trigger events have different requirements. There is no reasonable conflict resolution solution for these requirements yet.

SUMMARY

(5) Embodiments of the present disclosure provide a power control method and apparatus for direct communications, a terminal, and a storage medium, capable of determining whether a conflict on a 2-step RACH is resolved based on whether a message B meets a condition corresponding to a trigger event. Technical solutions are provided as follows.

(6) According to an aspect of the present disclosure, a 2-step random access method is provided. The method includes: transmitting, by a UE, a message A, when a 2-step random access is triggered based on a trigger event. The message A carries a Cell-Radio Network Temporary Identifier (C-RNTI) of the UE. The method further includes: receiving, by the UE, a message B; and determining, by the UE, that a conflict of the 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event.

(7) According to an aspect of the present disclosure, a 2-step random access method is provided. The method includes: receiving, by an access network device, a message A, which is transmitted by a UE based on a trigger event; and generating, by the access network device, a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event. The method further includes transmitting, by the access network device, the message B.

(8) In at least one embodiment, the trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover. Said transmitting, by the access network device, the message B includes: transmitting, by the access network device, a PDCCH scrambled by a C-RNTI, the PDCCH being used for scheduling downlink transmission; and transmitting, by the access network device, a Time Alignment Command (TAC) and an uplink scheduling grant based on the PDCCH.

(9) According to an aspect of the present disclosure, a 2-step random access apparatus is provided. The apparatus includes a transmitting module, a receiving module and a processing module. The transmitting module is configured to transmit a message a based on a trigger cause. The message A

carries a C-RNTI of a UE. The receiving module is configured to receive a message B. The processing module is configured to determine that a conflict of a 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event.

(10) According to an aspect of the present disclosure, a 2-step random access apparatus is provided. The apparatus includes a receiving module, a processing module and a transmitting module. The receiving module is configured to receive a message A. The message A is transmitted by a UE based on a trigger event, and carries a C-RNTI of the UE. The processing module is configured to generate a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event. The transmitting module is configured to transmit the message B. The message B indicates that a conflict of a 2-step random access is resolved, when a condition corresponding to the trigger event is satisfied.

(11) According to an aspect of the present disclosure, a communication terminal is provided. The communication terminal includes a processor, and a transceiver connected to the processor. The processor is configured to load and execute executable instructions to implement the 2-step random access method as described in any of the above aspects.

(12) According to an aspect of the present disclosure, a communication device is provided. The communication device includes a processor, and a transceiver connected to the processor. The processor is configured to load and execute executable instructions to implement the 2-step random access method as described in any of the above aspects.

(13) According to an aspect of the present disclosure, a computer-readable storage medium is provided. The computer-readable storage medium stores at least one instruction, at least one segment of a program, a set of codes, or a set of instructions. The at least one instruction, the at least one segment of the program, the set of codes, or the set of instructions is loaded and executed by a processor to implement the 2-step random access method as described in any of the above aspects.

(14) According to an aspect of the present disclosure, a chip is provided. The chip includes a programmable logic circuit and/or program instructions. When running, the chip is configured to implement the 2-step random access method as described in any of the above aspects.

(15) According to an aspect of the present disclosure, a computer program product is provided. The computer program product includes one or more computer programs. The one or more computer programs, when executed by a processor, implement the 2-step random access method as described in any of the above aspects.

(16) The technical solutions provided by the present disclosure can at least provide the following technical effects.

(17) Based on whether the message B satisfies the condition corresponding to a trigger event, it can be determined whether a conflict on a 2-step random access channel is resolved. In this way, for different trigger events, the UE can determine, based on different conflict resolution conditions, whether a conflict is resolved, thereby improving an access success rate of a 2-step random access process. A problem that the UE has different requirements in scenarios corresponding to various trigger events is solved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to clearly explain technical solutions in embodiments of the present disclosure, drawings used in the description of the embodiments are briefly described below. Obviously, the drawings as described below are merely some embodiments of the present disclosure. Based on these drawings, other drawings can be obtained by those skilled in the art without creative effort.

(2) FIG. 1 is a schematic diagram showing a communication system according to an exemplary

embodiment of the present disclosure.

(3) FIG. 2 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(4) FIG. 3 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(5) FIG. 4 is a format diagram of a Media Access Control Control Element (MAC CE) according to an exemplary embodiment of the present disclosure.

(6) FIG. 5 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(7) FIG. 6 is a format diagram of an MAC CE according to an exemplary embodiment of the present disclosure.

(8) FIG. 7 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(9) FIG. 8 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(10) FIG. 9 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(11) FIG. 10 is a format diagram of an MAC CE according to an exemplary embodiment of the present disclosure.

(12) FIG. 11 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure.

(13) FIG. 12 is a block diagram showing a structure of a 2-step random access apparatus according to an exemplary embodiment of the present disclosure.

(14) FIG. 13 is a block diagram showing a structure of a 2-step random access apparatus according to an exemplary embodiment of the present disclosure.

(15) FIG. 14 is a schematic block diagram of a communication terminal (a terminal or an access network device) according to an exemplary embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

(16) Exemplary embodiments will be described in detail here, and examples thereof are shown in the accompanying drawings. The same numbers in different drawings as described below indicate the same or similar elements, unless otherwise specified. Implementations described in the following exemplary embodiments, instead of representing all implementations consistent with the present disclosure, are merely examples of apparatuses and methods consistent with some aspects of the present disclosure as detailed in claims as attached.

(17) An RACH is an important channel in an initial access process between an access network device and a UE. LTE adopts a contention-based 4-step random access mechanism, which will be simplified to a contention-based 2-step random access mechanism in some usage scenarios of an NR system.

(18) Regarding the contention-based 2-step random access mechanism, the UE directly transmits a random access preamble and identity information to the access network device, and the access network device transmits an approach for resolving a conflict on the RACH along with a PDCCH scrambled by an RA-RNTI to the UE, so as to resolve the conflict on the RACH.

(19) In the 2-step random access mechanism, however, different trigger events have different requirements for conflict resolution. Regarding how to determine that a conflict in a 2-step random access is resolved under different trigger events, the present disclosure provides the following embodiments.

(20) FIG. 1 is a schematic diagram showing a communication system according to an exemplary embodiment of the present disclosure. An access network 12 includes a number of access network devices 120. The access network device 120 may be a base station. The base station is a device deployed in the access network to provide wireless communication functions for terminals. The

base station includes any form of a macro base station, a micro base station, a relay station, an access point, etc. In systems using different wireless access technologies, names of devices functioning as base stations may be different. For example, in an LTE system, a device functioning as a base station can be referred to as generation NodeB (gNB) or evolved NB (eNB); and in a 5th-Generation (5G) NR-Unlicensed (NR-U) system, the device functioning as a base station can be referred to as gNodeB or gNB. As communication technology develops, the phrase “base station” may change. For convenience, in the embodiments of the present disclosure, the devices described above that provide wireless communication functions for a terminal **13** are collectively referred to as the access network device.

(21) The terminal **13** may include various handheld devices, vehicle-mounted devices, wearable devices, computing devices with wireless communication functions or other processing devices connected to a wireless modem, and various forms of UE, Mobile Station (MS), terminal device, and the like. For the convenience of description, the devices mentioned above are collectively referred to as the terminal.

(22) In a communication process between the terminal and the access network device, once the terminal finds a cell, the terminal may access the cell by connecting to the access network device of the cell. A process for the terminal to connect to the access network device is referred to as a random access process. In the random access process, the access network device needs to assign to the terminal a unique identifier in the cell. In an actual scenario, a number of terminals may respond to the response message, and thus a conflict occurs. To resolve such a conflict, the access network device needs to assign a unique C-RNTI to the terminal.

(23) FIG. 2 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(24) At block **201**, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The message A carries a C-RNTI of the UE.

(25) A 2-step random access mechanism is an evolution of a 4-step random access mechanism. In a process of the 2-step random access mechanism, the UE can directly transmit a random access preamble and identity information (such as the C-RNTI) to the access network device.

(26) The message A includes the random access preamble and the identity information. That is, the message A includes contents of a message 1 and a message 3 in the 4-step random access.

(27) At block **202**, the access network device receives the message A.

(28) After the UE transmits the random access preamble and the identity information to the access network device, the access network device can receive the message A from the UE.

(29) At block **203**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(30) The message B carries a random access response and a message for conflict resolution, e.g., PDCCH information scrambled by the C-RNTI.

(31) At block **204**, the access network device transmits the message B.

(32) At block **205**, the UE receives the message B.

(33) After the UE transmits the message A, a detection window for the message B is opened.

(34) In at least embodiment, when it receives the message B, the UE may receive other information based on an instruction from the message B.

(35) In at least embodiment, when the UE receives the message B, the UE can determine whether the conflict is resolved.

(36) At block **206**, when the message B satisfies a condition corresponding to the trigger event, it is determined that a conflict of the 2-step random access is resolved.

(37) The condition corresponding to the trigger event is a conflict resolution condition corresponding to the trigger event. There are at least two trigger events that correspond to different

conflict resolution conditions.

(38) In summary, with the 2-step random access method according to the embodiment, it can be determined whether the conflict on the 2-step random access channel is resolved based on whether the message B satisfies the condition corresponding to a trigger event. In this way, for different trigger events, the UE can determine, based on different conflict resolution conditions, whether a conflict is resolved, thereby improving an access success rate of a 2-step random access process. A problem that the UE has different requirements in scenarios corresponding to various trigger events is solved.

(39) In a case that the message A includes a C-RNTI MAC CE, trigger events that trigger the UE to initiate the 2-step random access include at least the following seven types of trigger events.

(40) Trigger event 1 (abbreviated as E1)—an uplink data arrival and the UE being in an out-of-synchronization state.

(41) Trigger event 2 (abbreviated as E2)—an uplink data arrival but no Physical Uplink Control Channel (PUCCH) resource corresponding to a triggered Scheduling Request (SR).

(42) Trigger event 3 (abbreviated as E3)—SR transmission reaching a failure threshold.

(43) Trigger event 4 (abbreviated as E4)—a cell handover.

(44) Trigger event 5 (abbreviated as E5)—an addition of time synchronization for a secondary cell.

(45) Trigger event 6 (abbreviated as E6)—a beam failure recovery.

(46) Trigger event 7 (abbreviated as E7)—a downlink data arrival and the UE being in an uplink out-of-synchronization state.

(47) Among these trigger events, both the trigger event 1 and the trigger event 4 require the UE to obtain uplink TA on the one hand and an uplink scheduling grant on the other hand; in the trigger event 2 and the trigger event 3, since the UE is already in uplink synchronization, the uplink TA is unnecessary, but the uplink scheduling grant is still needed to transmit uplink data; in the trigger event 5 and the trigger event 7, the UE does not have uplink transmission to be scheduled, and only needs to obtain the uplink TA; and in the trigger event 6, the UE needs neither the uplink TA nor the uplink scheduling grant. In summary, the above seven trigger events can be summarized in the following table.

(48) TABLE-US-00001 TABLE 1 Trigger events that Trigger events that Trigger events that only need to obtain Trigger events that need to obtain both need neither the uplink the uplink only need to obtain the uplink TA and the TA nor the uplink scheduling grant the uplink TA uplink scheduling grant scheduling grant Uplink data arrival Addition of the time Uplink data arrival and Beam failure recovery but no PUCCH synchronization for the UE being in an out- resource the secondary cell of-synchronization state corresponding to the triggered SR SR transmission Downlink data arrival Cell handover reaching the failure and the UE being in threshold an uplink out-of-synchronization state

(49) The following embodiments describe the above four cases.

(50) Case 1

(51) FIG. 3 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(52) At block 301, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover.

(53) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a Physical Random Access Channel (PRACH) resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on a Physical Uplink Shared Channel (PUSCH) for transmission.

(54) The 2-step random access is triggered by the E1 or the E4. That is, the 2-step random access is

triggered by a MAC sublayer or a Radio Resource Control (RRC) sublayer. In at least embodiment, the trigger cause is the UE being in the uplink out-of-synchronization state or the cell handover.

(55) At block **302**, the access network device receives the message A.

(56) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(57) At block **303**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(58) When receiving the message A, the access network device learns other information (such as context information, state information of a synchronization timer, etc.) of the UE based on the C-RNTI in the message A, and learns the trigger event of the message A based on the other information and the state of the UE.

(59) In this embodiment, since the trigger event of the message A is the UE being in the uplink out-of-synchronization state or the cell handover, the message B needs to meet a conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(60) In at least embodiment, the message B may directly include conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict, or downlink transmission (or uplink transmission) scheduled by information in the message B may include the conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict.

(61) At block **304**, the access network device transmits the message B. The message B includes the PDCCH scrambled by the C-RNTI, and the PDCCH is used for scheduling downlink transmission.

(62) In at least embodiment, the PDCCH carries Downlink Control Information (DCI). The DCI is used for scheduling the downlink transmission.

(63) At block **305**, the UE receives the message B.

(64) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(65) After receiving the message B in the detection window, the UE obtains the DCI in the PDCCH.

(66) At block **306**, the access network device transmits a first MAC Protocol Data Unit (PDU) based on the downlink transmission scheduled by the PDCCH. The first MAC PDU carries a first MAC CE, and the first MAC CE carries a Time Alignment Command (TAC) and an uplink scheduling grant (UL grant for short).

(67) FIG. 4 is a format diagram of a first MAC CE according to an exemplary embodiment of the present disclosure. Referring to FIG. 4, in a message format of the first MAC CE, the first MAC CE occupies 5 bytes, and each byte has 8 bits. The TAC occupies 12 bits in byte 1 and byte 2, and the uplink scheduling grant occupies 27 bits in byte 2 to byte 5. Here, R is a reserved bit.

(68) In at least embodiment, numbers of bits occupied by the TAC and the uplink scheduling grant may be different from those in FIG. 4, which are not limited in this embodiment.

(69) The TAC is used to enable the UE to regain uplink synchronization with the access network device. The uplink scheduling grant is used for scheduling resources for the UE to perform uplink transmission.

(70) In this embodiment, the TAC and the uplink scheduling grant are packaged in a same MAC CE for transmission. The TAC and the uplink scheduling grant correspond to a same MAC subheader, and only need one Logical Channel Identifier (LCID) indication.

(71) At block **307**, the UE decodes the first MAC PDU based on the downlink transmission scheduled by the PDCCH. The first MAC PDU carries the first MAC CE.

(72) The UE receives and decodes the first MAC PDU based on scheduling of the DCI in the PDCCH, and the first MAC PDU carries the first MAC CE.

(73) At block **308**, when the first MAC CE carries the TAC and the uplink scheduling grant, it is

determined that the conflict of the 2-step random access is resolved.

(74) In summary, with the 2-step random access method according to this embodiment, when the UE needs to obtain both the uplink TA and the uplink scheduling grant, the method of using the PDCCH to schedule a single MAC CE that includes both the TAC and the uplink scheduling grant enables transmission of conflict resolution to be completed with only one MAC CE, which simplifies and reduces interaction procedures between the UE and the access network device.

(75) FIG. 5 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(76) At block **501**, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover.

(77) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a PRACH resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on the PUSCH for transmission.

(78) The 2-step random access is triggered by the E1 or the E4. That is, the 2-step random access is triggered by the MAC sublayer or the RRC sublayer. In at least embodiment, the message A also carries the trigger cause, i.e., the UE being in the uplink out-of-synchronization state, or the cell handover.

(79) At block **502**, the access network device receives the message A.

(80) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(81) At block **503**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(82) When receiving the message A, the access network device learns other information (such as context information, state information of a synchronization timer, etc.) of the UE based on the C-RNTI in the message A, and learns the trigger event of the message A based on the other information and the state of the UE.

(83) In this embodiment, since the trigger event of the message A is the UE being in the uplink out-of-synchronization state or the cell handover, the message B needs to meet a conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(84) In at least embodiment, the message B may directly include conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict, or downlink transmission (or uplink transmission) scheduled by information in the message B may include the conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict.

(85) At block **504**, the access network device transmits the message B. The message B includes the PDCCH scrambled by the C-RNTI, and the PDCCH is used for scheduling downlink transmission.

(86) In at least embodiment, the PDCCH carries the DCI. The DCI is used for scheduling the downlink transmission.

(87) At block **505**, the UE receives the message B.

(88) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(89) After receiving the message B in the detection window, the UE obtains the DCI in the PDCCH.

(90) At block **506**, the access network device transmits a second MAC PDU based on the downlink transmission scheduled by the PDCCH. The second MAC PDU carries a second MAC CE and a third MAC CE, the second MAC CE carries the TAC, and the third MAC CE carries the uplink

scheduling grant.

(91) FIG. 6 is a format diagram of a second MAC CE and a third MAC CE according to an exemplary embodiment of the present disclosure. Referring to (a) in FIG. 6, in a format of the second MAC CE, the second MAC CE occupies 16 bits in byte 1 and byte 2. Referring to (b) in FIG. 6, in a format of the third MAC CE, the third MAC CE occupies 32 bits in byte 1 to byte 4. Here, R is a reserved bit.

(92) In at least embodiment, numbers of bits occupied by the TAC and the uplink scheduling grant may be different from those in FIG. 4, which are not limited in this embodiment.

(93) The TAC is used to enable the UE to regain uplink synchronization with the access network device. The uplink scheduling grant is used for scheduling resources for the UE to perform uplink transmission.

(94) In this embodiment, the TAC and the uplink scheduling grant are packaged in different MAC CEs for transmission. The TAC and the uplink scheduling grant correspond to different MAC subheaders, and the TAC and the uplink scheduling grant each require an LCID indication. That is, the TAC and the uplink scheduling grant have different LCID indications.

(95) At block 507, the UE decodes a second MAC PDU based on the downlink transmission scheduled by the PDCCH. The second MAC PDU carries a second MAC CE and a third MAC CE.

(96) The UE decodes the second MAC CE including the TAC and the third MAC CE including the uplink scheduling grant from the second MAC PDU based on the downlink transmission scheduled by the PDCCH.

(97) At block 508, when the second MAC CE carries the TAC, and the third MAC CE carries the uplink scheduling grant, it is determined that the conflict of the 2-step random access is resolved.

(98) After obtaining the TAC and the uplink scheduling grant by decoding the two MAC CEs separately, the UE can determine that the conflict of the 2-step random access is resolved.

(99) In summary, with the 2-step random access method according to this embodiment, when the UE needs to obtain both the uplink TA and the uplink scheduling grant, the method of using the PDCCH to schedule two MAC CEs that include both the TAC and the uplink scheduling grant enables the TAC and the uplink scheduling grant to be transmitted separately, which increases flexibility of interaction between the UE and the access network device.

(100) FIG. 7 is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(101) At block 701, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover.

(102) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a PRACH resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on the PUSCH for transmission.

(103) The 2-step random access is triggered by the E1 or the E4. That is, the 2-step random access is triggered by the MAC sublayer or the RRC sublayer. In at least embodiment, the message A also carries the trigger cause, i.e., the UE being in the uplink out-of-synchronization state, or the cell handover.

(104) At block 702, the access network device receives the message A.

(105) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(106) At block 703, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(107) When receiving the message A, the access network device learns other information (such as

context information, state information of a synchronization timer, etc.) of the UE based on the C-RNTI in the message A, and learns the trigger event of the message A based on the other information and the state of the UE.

(108) In this embodiment, since the trigger event of the message A is the UE being in the uplink out-of-synchronization state or the cell handover, the message B needs to meet a conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(109) In at least embodiment, the message B may directly include conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict, or downlink transmission (or uplink transmission) scheduled by information in the message B may include the conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict.

(110) At block **704**, the access network device transmits the message B. The message B includes the PDCCH scrambled by the C-RNTI. DCI is transmitted in the PDCCH. The DCI carries the TAC and the uplink scheduling grant.

(111) In at least embodiment, the PDCCH carries the DCI, and the DCI carries the TAC and the uplink scheduling grant. The DCI is used for scheduling the uplink transmission. The uplink scheduling grant is used for new data transmission.

(112) At block **705**, the UE receives the message B.

(113) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(114) At block **706**, when the DCI carries the TAC and the uplink scheduling grant, it is determined that a conflict of a 2-step random access is resolved.

(115) When the DCI carries the TAC and the uplink scheduling grant, it is determined that the conflict of the 2-step random access is resolved.

(116) In summary, with the 2-step random access method according to this embodiment, the TAC and the uplink scheduling grant can be obtained after the UE receives the PDCCH of the message B, such that the UE can learn that the conflict is resolved as quickly as possible, thereby shortening time consumed by the 2-step random access.

(117) FIG. **8** is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(118) At block **801**, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes SR transmission reaching a failure threshold, or an uplink data arrival but no PUCCH resource corresponding to an SR triggered by uplink data.

(119) The SR refers to an uplink scheduling request of the UE. When uplink data needs to be transmitted, the UE needs to transmit an SR to the access network device.

(120) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a PRACH resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on the PUSCH for transmission.

(121) The 2-step random access is triggered by the E2 or the E3. In at least embodiment, the trigger cause is also carried in the message A.

(122) At block **802**, the access network device receives the message A.

(123) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(124) At block **803**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(125) When receiving the message A, the access network device learns other information (such as context information, state information of a synchronization timer, etc.) of the UE based on the C-

RNTI in the message A, and learns the trigger event of the message A based on the other information and the state of the UE.

(126) In this embodiment, since the trigger event of the message A is the SR transmission reaching the failure threshold, or the uplink data arrival but no PUCCH resource corresponding to the triggered SR, the message B needs to satisfy the conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(127) At block **804**, the access network device transmits the message B to the UE. The message B includes the PDCCH scrambled by the C-RNTI, and the PDCCH carries the uplink scheduling grant.

(128) In at least embodiment, the PDCCH carries the DCI, and the DCI is used for scheduling the uplink transmission. The DCI carries the uplink scheduling grant.

(129) At block **805**, the UE receives the message B.

(130) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(131) After the UE receives the message B in the detection window, the UE obtains the DCI in the PDCCH.

(132) At block **806**, when a PDCCH scrambled by the C-RNTI is detected in the detection window of the message B and the PDCCH has an uplink scheduling grant, it is determined that the conflict of the 2-step random access is resolved.

(133) It is determined that the conflict of the 2-step random access is resolved, when the PDCCH scrambled by the C-RNTI is detected in the detection window of the message B, and the DCI that carries the uplink scheduling grant is decoded from the PDCCH.

(134) In summary, with the 2-step random access method according to this embodiment, when the UE only needs to obtain the uplink scheduling grant, the uplink scheduling grant can be transmitted directly to the UE in the message B, which simplifies and reduces interaction processes between the UE and the access network device.

(135) FIG. **9** is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(136) At block **901**, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes an addition of time synchronization for a secondary cell, or a downlink data arrival and the UE being in an uplink out-of-synchronization state.

(137) When the UE is in a primary cell and needs to perform information transmission with the access network device of the secondary cell, time synchronization with the secondary cell needs to be performed. In this case, the UE needs to obtain the uplink TA. Or, when a downlink data arrival occurs at the UE and the UE is in the uplink out-of-synchronization state, the UE also needs to obtain the uplink TA.

(138) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a PRACH resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on the PUSCH for transmission.

(139) The 2-step random access is triggered by the E5 or the E7. In at least embodiment, the trigger cause is also carried in the message A.

(140) At block **902**, the access network device receives the message A.

(141) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(142) At block **903**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(143) When receiving the message A, the access network device learns the trigger event of the message A based on information in the message A and other information and states of the UE and the access network device.

(144) In this embodiment, since the trigger event of the message A is the addition of the time synchronization for the secondary cell, or the downlink data arrival and the UE being in the uplink out-of-synchronization state, the message B needs to satisfy the conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(145) At block **904**, the access network device transmits the message B to the UE. The message B includes the PDCCH scrambled by the C-RNTI, and the PDCCH is used for scheduling the downlink transmission.

(146) In at least embodiment, the PDCCH carries the DCI, and the DCI is used for scheduling the downlink transmission.

(147) At block **905**, the UE receives the message B.

(148) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(149) After the UE receives the message B in the detection window, the UE obtains the DCI in the PDCCH.

(150) At block **906**, the access network device transmits a third MAC PDU to the UE based on the downlink transmission scheduled by the PDCCH. The third MAC PDU includes a fourth MAC CE, and the fourth MAC CE carries a TAC.

(151) FIG. **10** is a format diagram of a fourth MAC CE according to an exemplary embodiment of the present disclosure. Referring to FIG. **10**, in the format of the fourth MAC CE, the TAC occupies 12 bits of byte **1** and byte **2**.

(152) In at least embodiment, the number of bits occupied by the TAC may be different from that in FIG. **10**, and this embodiment of the present disclosure is not limiting in this regard.

(153) At block **907**, the UE decodes the third MAC PDU based on the downlink transmission scheduled by the PDCCH, the third MAC PDU including the fourth MAC CE, the fourth MAC CE carrying the TAC; and the UE determines that the conflict of the 2-step random access is resolved.

(154) In summary, with the 2-step random access method according to this embodiment, when the UE only needs to obtain the TAC, the TAC is transmitted to the UE via the downlink transmission scheduled by the PDCCH of the message B, which simplifies and reduces interaction processes between the UE and the access network device.

(155) FIG. **11** is a flowchart illustrating a 2-step random access method according to an exemplary embodiment of the present disclosure. The method may be executed by a UE and an access network device. The method includes the following operations.

(156) At block **1101**, when a 2-step random access is triggered based on a trigger event, the UE transmits a message A. The trigger event includes a beam failure recovery.

(157) The beam failure recovery is a function of beam management. When a base station beam and a terminal beam that are originally aligned with each other are blocked by an obstacle (such as a person, a vehicle, etc.), it is necessary to find a new pair of beams that can be aligned with each other on a certain reflection path to ensure that communication can continue.

(158) The message A includes a random access preamble and a payload. The random access preamble is transmitted on a PRACH resource. The payload carries the content of the message 3 for the 4-step random access. For example, the payload carries a MAC CE, and the MAC CE carries the C-RNTI of the UE. The payload is carried on the PUSCH for transmission.

(159) The 2-step random access is triggered by the E6. In at least embodiment, the trigger cause is also carried in the message A.

(160) At block **1102**, the access network device receives the message A.

(161) The access network device receives the random access preamble on the PRACH and receives the payload on the PUSCH.

(162) At block **1103**, the access network device generates a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(163) When receiving the message A, the access network device learns the trigger event of the message A based on the information in the message A and other information and states of the UE and the access network device.

(164) In this embodiment, since the trigger event of the message A is the beam failure recovery, the message B needs to satisfy the conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(165) At block **1104**, the access network device transmits the message B to the UE. The message B includes the PDCCH scrambled by the C-RNTI.

(166) In this embodiment, since the trigger event of the message A is the UE being in the uplink out-of-synchronization state or the cell handover, the message B needs to satisfy the conflict resolution condition corresponding to the trigger event of the message A that can resolve the conflict.

(167) In at least embodiment, the message B may directly include conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict, or the downlink transmission (or the uplink transmission) scheduled by the information in the message B may include the conflict resolution information corresponding to the trigger event of the message A that can resolve the conflict.

(168) At block **1105**, the UE receives the message B.

(169) After the UE transmits the message A, the detection window for the message B is opened. The UE uses its own C-RNTI to descramble the PDCCH.

(170) At block **1106**, when the PDCCH scrambled by the C-RNTI is received in the detection window of the message B, it is determined that the conflict of the 2-step random access is resolved.

(171) In summary, with the 2-step random access method according to this embodiment, in a scenario of the beam failure recovery, the conflict can be considered to be successfully resolved when the access network device transmits the PDCCH scrambled by the C-RNTI to the UE, which simplifies and reduces interaction processes between the UE and the access network device.

(172) Based on the description of the above embodiments, during transmission of the message A, an entity at a MAC layer performs the following procedures:

(173) 1> At the end of the transmission of the message A, the detection window for the message B is opened in a 1-st PDCCH.

(174) 1> During an operation of the detection window for the message B, the message B in the primary cell and the secondary cell in the PDCCH is monitored through the RA-RNTI and the C-RNTI.

(175) 1> If a reception notification in the PDCCH in the primary cell or the secondary cell is obtained from a lower layer, then:

(176) 2> if a C-RNTI MAC CE is included in the message A, then:

(177) 3> when a random access procedure starts from a beam failure process and an address of the PDCCH transmission is located by the C-RNTI, or

(178) 3> when the random access procedure starts with a PDCCH instruction, and the MAC PDU includes a MAC CE carrying a TAC, or

(179) 3> when the random access procedure starts from a MAC sublayer (for example, the uplink data arrival but no PUCCH resource corresponding to the triggered SR, or the SR transmission reaching the failure threshold), and a PDCCH whose transmission address is located to the C-RNTI includes an uplink scheduling grant corresponding to a new transmission, or

(180) 3> when the uplink data arrival occurs and the UE is in an out-of-synchronization state (for example, for scene switching) and the PDCCH is located to the C-RNTI, and the MAC PDU is successfully decoded and the MAC PDU includes a MAC CE of the TAC and the uplink

scheduling grant, then:

(181) 4> it is determined that a correct solution is provided;

(182) 4> the operation of the detection window for the message B is stopped; and

(183) 4> it is determined that the random access process is successfully completed.

(184) FIG. 12 is a block diagram showing a structure of a 2-step random access apparatus according to an exemplary embodiment of the present disclosure. The apparatus can be implemented as the UE or part of the UE via software, hardware or a combination thereof. The apparatus includes a receiving module **1201**, a processing module **1202**, and a transmitting module **1203**.

(185) The transmitting module **1203** is configured to transmit a message A when a 2-step random access is triggered based on a trigger event. The message A carries a C-RNTI of a UE.

(186) The receiving module **1201** is configured to receive a message B.

(187) The processing module **1202** is configured to determine that a conflict of the 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event.

(188) In an example, the trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover.

(189) The processing module **1202** being configured to determine that the conflict of the 2-step random access is resolved, when the message B satisfies the condition corresponding to the trigger event, includes: the processing module **1202** being configured to detect a PDCCH scrambled by the C-RNTI in a detection window of the message B; and the receiving module **1201** being configured to receive a TAC and an uplink scheduling grant based on the PDCCH, and to determine that the conflict of the 2-step random access is resolved.

(190) In an example, the PDCCH is used for scheduling downlink transmission.

(191) The receiving module **1201** being configured to receive the TAC and the uplink scheduling grant based on the PDCCH, and to determine that the conflict of the 2-step random access channel is resolved includes: decoding, by the UE, a MAC PDU based on the downlink transmission scheduled by the PDCCH, the MAC PDU including a first MAC CE, the first MAC CE carrying the TAC and the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved. Or it includes: decoding, by the UE, a MAC PDU based on the downlink transmission scheduled by the PDCCH, the MAC PDU including a second MAC CE and a third MAC CE, the second MAC CE carrying the TAC, the third MAC CE carrying the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved, the uplink scheduling grant being used for new data transmission.

(192) In an example, the PDCCH is used for scheduling uplink transmission.

(193) The processing module **1202** is configured to determine, when DCI decoded based on the PDCCH carries the TAC and the uplink scheduling grant, that the conflict on the random access channel is resolved. The uplink scheduling grant is used for new data transmission.

(194) In an example, the trigger event includes SR transmission reaching a failure threshold, or an uplink data arrival but no PUCCH resource corresponding to an SR triggered by uplink data.

(195) The processing module **1202** is configured to determine that the conflict of the 2-step random access is resolved, when a PDCCH scrambled by the C-RNTI is detected in a detection window of the message B and the PDCCH carries an uplink scheduling grant.

(196) In an example, the trigger event includes an addition of time synchronization for a secondary cell, or a downlink data arrival and the UE being in an uplink out-of-synchronization state.

(197) The processing module **1202** is configured to detect a PDCCH scrambled by the C-RNTI in a detection window of the message B.

(198) The processing module **1202** is configured to decode a MAC PDU by the UE based on the downlink transmission scheduled by the PDCCH, the MAC PDU including a fourth MAC CE, the fourth MAC CE carrying a TAC; and to determine that the conflict of the 2-step random access is resolved.

(199) In an example, the trigger event includes a beam failure recovery.

(200) The processing module **1202** is configured to determine that the conflict on the 2-step random access is resolved, when a PDCCH scrambled by the C-RNTI is detected in a detection window of the message B.

(201) FIG. **13** is a block diagram showing a structure of a 2-step random access apparatus according to an exemplary embodiment of the present disclosure. The apparatus can be implemented as the access network device or part of the access network device via software, hardware or a combination thereof. The apparatus includes a receiving module **1301**, a processing module **1302**, and a transmitting module **1303**.

(202) The receiving module **1301** is configured to receive a message A. The message A is transmitted by a UE based on a trigger event.

(203) The processing module **1302** is configured to generate a message B based on the trigger event of the message A. The message B satisfies a conflict resolution condition corresponding to the trigger event.

(204) The transmitting module **1303** is configured to transmit the message B.

(205) In an example, the trigger event includes an uplink data arrival and the UE being in an uplink out-of-synchronization state, or a cell handover.

(206) The transmitting module **1303** is configured to transmit by the access network device a PDCCH scrambled by the C-RNTI. The PDCCH is used for scheduling downlink transmission.

(207) The transmitting module is configured to transmit by the access network device a TAC and an uplink scheduling grant based on the PDCCH.

(208) In an example, the transmitting module **1303** is configured to transmit a first MAC CE based on the downlink transmission scheduled by the PDCCH, the first MAC CE carrying the TAC and the uplink scheduling grant. Or the access network device is configured to transmit a second MAC CE and a third MAC CE based on the downlink transmission scheduled by the PDCCH, the second MAC CE carrying the TAC, the third MAC CE carrying the uplink scheduling grant.

(209) In an example, the transmitting module **1303** is configured to transmit DCI in the PDCCH. The DCI carries the TAC and the uplink scheduling grant.

(210) In an example, the trigger event includes SR transmission reaching a failure threshold, or an uplink data arrival but no PUCCH resource corresponding to an SR triggered by uplink data.

(211) The transmitting module **1303** is configured to transmit a PDCCH scrambled by the C-RNTI. The PDCCH carries an uplink scheduling grant.

(212) In an example, the trigger event includes an addition of time synchronization for a secondary cell, or a downlink data arrival and the UE being in an uplink out-of-synchronization state.

(213) The transmitting module **1303** is configured to transmit a PDCCH scrambled by the C-RNTI, the PDCCH being used for scheduling downlink transmission; and to transmit a TAC based on the downlink transmission scheduled by the PDCCH.

(214) In an example, the trigger event includes a beam failure recovery.

(215) The transmitting module **1303** is configured to transmit a PDCCH scrambled by the C-RNTI.

(216) FIG. **14** is a schematic block diagram of a communication terminal (a terminal or an access network device) according to an exemplary embodiment of the present disclosure. The communication terminal includes a processor **1401**, a receiver **1402**, a transmitter **1403**, a memory **1404**, and a bus **1405**.

(217) The processor **1401** includes one or more processing cores. The processor **1401** executes various functional applications and information processing by running software programs and modules.

(218) The receiver **1402** and the transmitter **1403** can be implemented as one communication component. The communication component can be a communication chip.

(219) The memory **1404** is connected to the processor **1405** via the bus **1401**.

(220) The memory **1404** can be used to store at least one instruction. The processor **1401** is

configured to execute the at least one instruction, so as to implement each operation in the above method embodiments.

(221) In addition, the memory **1404** may be implemented by any type of volatile or non-volatile storage device or a combination thereof. The volatile or non-volatile storage device includes, but is not limited to, a magnetic disk or an optical disk, an Electrically Erasable Programmable Read-Only Memory (EEPROM), an Erasable Programmable Read-Only Memory (EPROM), a Static Random Access Memory (SRAM), a Read-Only Memory (ROM), a magnetic memory, a flash memory, and a Programmable Read-Only Memory (PROM).

(222) In an exemplary embodiment, a computer-readable storage medium is provided. The computer-readable storage medium stores at least one instruction, at least one segment of a program, a set of codes, or a set of instructions. The at least one instruction, the at least one segment of the program, the set of codes, or the set of instructions is loaded and executed by a processor to implement the method for transmitting/receiving uplink data performed by a communication device according to any of the above method embodiments.

(223) In an exemplary embodiment, a computer-readable storage medium is provided. The computer-readable storage medium stores at least one instruction, at least one segment of a program, a set of codes, or a set of instructions. The at least one instruction, the at least one segment of the program, the set of codes, or the set of instructions is loaded and executed by a processor to implement the power control method of direct communications according to any of the above method embodiments.

(224) It should be understood that the term “a plurality of” mentioned herein refers to two or more. The term “and/or” describes an association relationship between correlated objects, including three relationships. For example, “A and/or B” may mean three situations: A only, B only, or both A and B. In addition, the character “/” generally represents an “or” relationship between the correlated objects preceding and succeeding the symbol.

(225) Other implementations of the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the present disclosure disclosed here. The present disclosure is intended to cover any variations, uses, or adaptations of the present disclosure following the general principles thereof and including such departures from the present disclosure as come within known or customary technical means in the art. The specification and embodiments are considered as exemplary only, with a true scope and spirit of the present disclosure being indicated by the claims as attached.

(226) It can be appreciated that the present disclosure is not limited to the exact structure that has been described above and illustrated in the accompanying drawings, and that various modifications and changes can be made without departing from the scope of the present disclosure. The scope of the present disclosure is only limited by the claims as attached.

Claims

1. A 2-step random access method, comprising: transmitting, by a User Equipment (UE), a message A, when a 2-step random access is triggered based on a trigger event, wherein the message A carries a Cell-Radio Network Temporary Identifier (C-RNTI) of the UE; receiving, by the UE, a message B; and determining, by the UE, that a conflict of the 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event, wherein the trigger event comprises: Scheduling Request (SR) transmission reaching a failure threshold, and said determining that the conflict of the 2-step random access is resolved, when the message B satisfies the condition corresponding to the trigger event comprises: determining that the conflict of the 2-step random access is resolved, when the UE detects a Physical Downlink Control Channel (PDCCH) scrambled by the C-RNTI in a detection window of the message B and the PDCCH carries an uplink scheduling grant, or wherein the trigger event comprises: an addition of time

synchronization for a secondary cell, and the PDCCH is used for scheduling downlink transmission, and said determining that the conflict of the 2-step random access is resolved, when the message B satisfies the condition corresponding to the trigger event comprises: detecting, by the UE, a PDCCH scrambled by the C-RNTI in a detection window of the message B; and decoding, by the UE, a third Media Access Control Protocol Data Unit (MAC PDU) based on the downlink transmission scheduled by the PDCCH, wherein the third MAC PDU comprises a fourth MAC Control Element (CE), and the fourth MAC CE carries a Time Alignment Command (TAC); and determining that the conflict of the 2-step random access is resolved.

2. The method according to claim 1, wherein the PDCCH is used for scheduling downlink transmission, and said receiving the TAC and the uplink scheduling grant based on the PDCCH, and determining that the conflict of the 2-step random access is resolved comprises: decoding, by the UE, a first MAC PDU based on the downlink transmission scheduled by the PDCCH, wherein the first MAC PDU comprises a first MAC CE, and the first MAC CE carries the TAC and the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved; or decoding, by the UE, a second MAC PDU based on the downlink transmission scheduled by the PDCCH, wherein the second MAC PDU comprises a second MAC CE and a third MAC CE, the second MAC CE carries the TAC, and the third MAC CE carries the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved, wherein the uplink scheduling grant is used for new data transmission.

3. The method according to claim 1, wherein the PDCCH is used for scheduling uplink transmission, and said receiving the TAC and the uplink scheduling grant based on the PDCCH, and determining that the conflict of the 2-step random access is resolved comprises: determining, when Downlink Control Information (DCI) decoded by the UE based on the PDCCH carries the TAC and the uplink scheduling grant, that a conflict on a random access channel is resolved, wherein the uplink scheduling grant is used for new data transmission.

4. A 2-step random access method, comprising: receiving, by an access network device, a message A, wherein the message A is transmitted by a User Equipment (UE) based on a trigger event; generating, by the access network device, a message B based on the trigger event of the message A, wherein the message B satisfies a conflict resolution condition corresponding to the trigger event; and transmitting, by the access network device, the message B, wherein the trigger event comprises: Scheduling Request (SR) reaching a failure threshold, and said transmitting, by the access network device, the message B comprises: transmitting a Physical Downlink Control Channel (PDCCH) scrambled by a Cell-Radio Network Temporary Identifier (C-RNTI), wherein the PDCCH carries an uplink scheduling grant, or wherein the trigger event comprises: an addition of time synchronization for a secondary cell, and said transmitting, by the access network device, the message B comprises: transmitting a PDCCH scrambled by a Cell-Radio Network Temporary Identifier (C-RNTI), wherein the PDCCH is used for scheduling downlink transmission; and transmitting a third Media Access Control Protocol Data Unit (MAC PDU) based on the downlink transmission scheduled by the PDCCH, wherein the third MAC PDU comprises a fourth MAC Control Element (CE), and the fourth MAC CE carries a Time Alignment Command (TAC).

5. The method according to claim 4, wherein the method further comprises: transmitting, by the access network device, a first MAC CE based on the downlink transmission scheduled by the PDCCH, the first MAC CE carrying the TAC and the uplink scheduling grant, or transmitting, by the access network device, a second MAC CE and a third MAC CE based on the downlink transmission scheduled by the PDCCH, the second MAC CE carrying the TAC, the third MAC CE carrying the uplink scheduling grant.

6. The method according to claim 4, wherein the method further comprises: transmitting, by the access network device, DCI in the PDCCH, wherein the DCI carries the TAC and the uplink scheduling grant.

7. A communication terminal, comprising: a processor; and a transceiver connected to the

processor, wherein the processor is configured to load and execute executable instructions to implement a 2-step random access method comprising: transmitting a message A, when a 2-step random access is triggered based on a trigger event, wherein the message A carries a Cell-Radio Network Temporary Identifier (C-RNTI) of the UE; receiving a message B; and determining that a conflict of the 2-step random access is resolved, when the message B satisfies a condition corresponding to the trigger event, wherein the trigger event comprises: Scheduling Request (SR) transmission reaching a failure threshold, and said determining that the conflict of the 2-step random access is resolved, when the message B satisfies the condition corresponding to the trigger event comprises: determining that the conflict of the 2-step random access is resolved, when the UE detects a Physical Downlink Control Channel (PDCCH) scrambled by the C-RNTI in a detection window of the message B and the PDCCH carries an uplink scheduling grant, or wherein the trigger event comprises: an addition of time synchronization for a secondary cell, and the PDCCH is used for scheduling downlink transmission, and said determining that the conflict of the 2-step random access is resolved, when the message B satisfies the condition corresponding to the trigger event comprises: detecting a PDCCH scrambled by the C-RNTI in a detection window of the message B; and decoding a third Media Access Control Protocol Data Unit (MAC PDU) based on the downlink transmission scheduled by the PDCCH, wherein the third MAC PDU comprises a fourth MAC Control Element (CE), and the fourth MAC CE carries a Time Alignment Command (TAC); and determining that the conflict of the 2-step random access is resolved.

8. The communication terminal according to claim 7, wherein the PDCCH is used for scheduling downlink transmission, and said receiving the TAC and the uplink scheduling grant based on the PDCCH, and determining that the conflict of the 2-step random access is resolved comprises: decoding a first MAC PDU based on the downlink transmission scheduled by the PDCCH, wherein the first MAC PDU comprises a first MAC CE, and the first MAC CE carries the TAC and the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved; or decoding a second MAC PDU based on the downlink transmission scheduled by the PDCCH, wherein the second MAC PDU comprises a second MAC CE and a third MAC CE, the second MAC CE carries the TAC, and the third MAC CE carries the uplink scheduling grant; and determining that the conflict of the 2-step random access is resolved, wherein the uplink scheduling grant is used for new data transmission.

9. The communication terminal according to claim 7, wherein the PDCCH is used for scheduling uplink transmission, and said receiving the TAC and the uplink scheduling grant based on the PDCCH, and determining that the conflict of the 2-step random access is resolved comprises: determining, when Downlink Control Information (DCI) decoded by the UE based on the PDCCH carries the TAC and the uplink scheduling grant, that a conflict on a random access channel is resolved, wherein the uplink scheduling grant is used for new data transmission.

10. A communication device, comprising: a processor; and a transceiver connected to the processor, wherein the processor is configured to load and execute executable instructions to implement the 2-step random access method according to claim 4.

11. The communication device according to claim 10, wherein the method further comprises: transmitting a first MAC CE based on the downlink transmission scheduled by the PDCCH, the first MAC CE carrying the TAC and the uplink scheduling grant, or transmitting a second MAC CE and a third MAC CE based on the downlink transmission scheduled by the PDCCH, the second MAC CE carrying the TAC, the third MAC CE carrying the uplink scheduling grant.

12. The communication device according to claim 10, wherein the method further comprises: transmitting DCI in the PDCCH, wherein the DCI carries the TAC and the uplink scheduling grant.
